

Pay-By-Face (dApp): A Decentralized Biometric Payment Application Integrating Face Recognition and Blockchain Smart Wallets

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Abstract: Modern financial checkout environments continue to depend heavily on tangible intermediary tokens, including plastic debit or credit cards, mobile phones, NFC transponders, and quick-response codes. These instruments frequently subject patrons to skimming risks, accidental loss, device power depletion, and credential theft. Conversely, distributed ledger infrastructures such as Ethereum deliver immutable transaction settlement without central clearing houses, yet remain hindered by severe usability friction, including the handling of 24-word recovery seeds and 42-character hexadecimal account addresses. To resolve this dilemma, this work introduces Pay-By-Face (dApp), a novel decentralized financial application uniting edge-computed facial computer vision with automated blockchain smart wallets. Operating in-browser via WebGL acceleration, the platform employs a Single Shot MultiBox Detector (SSD MobileNet v1), 68-point landmark alignment, and a residual network to extract 128-dimensional Euclidean feature vectors directly on user devices without sending raw images over the network. Identity determination is regulated by an empirical distance threshold of $T = 0.55$, supplemented by a secondary four-digit bcrypt-salted passcode to construct a dependable two-factor authentication mechanism against 2D presentation spoofing. Following dual verification, the serverless backend decrypts user keyrings in temporary memory using AES-256-GCM and broadcasts transactions over the Ethereum Sepolia network via Viem under EIP-1559 gas rules. Empirical assessments demonstrate an Equal Error Rate of 1.84%, an authentication latency of approximately 365 ms, and verifiable on-chain finality, proving that non-custodial biometric retail transactions can function seamlessly without centralized payment intermediaries.

Index Terms: Biometric Payments, Deep Metric Learning, 128-Dimensional Embeddings, Ethereum Blockchain, Decentralized Applications (dApps), Smart Wallets, Viem, Two-Factor Authentication, Non-Custodial Fintech, Point-of-Sale Systems.

I. INTRODUCTION

The trajectory of point-of-sale retail processing across recent decades reflects a continuous endeavor to eliminate user effort while preserving fiscal accounting rigor. Originating from physical fiat notes to magnetic stripe mechanisms, EMV integrated circuits, and near-field contactless transmissions via smartphones, each phase has lowered interface barriers. However, contemporary transaction pipelines remain tethered to external possessions (such as smart cards, fobs, and mobile devices) or memorized credentials (including PINs and one-time passwords). Consequently, cardholders face persistent risks from hardware loss, battery drain, SIM-swap hijacks, and optical snooping.

Physiological biometrics, representing innate human identifiers, present a compelling pathway toward device-independent payment settlement. Facial recognition offers a touchless and intuitive user interaction paradigm.

Nonetheless, conventional commercial implementations deployed by large technology conglomerates suffer from notable structural drawbacks: the concentration of raw biometric imagery in centralized repositories creates severe single-point-of-failure vulnerabilities, and closed-loop proprietary ledgers impose substantial processing surcharges while depriving consumers of sovereign asset possession.

In parallel, public smart contract networks like Ethereum provide decentralized value transfer, cryptographic non-repudiation, and programmatic composability. Unfortunately, decentralized applications (dApps) remain widely inaccessible to non-technical individuals due to daunting user experience hurdles. Consumers must safeguard complex mnemonic phrases, install specialized browser plugins, inspect lengthy hexadecimal strings, and estimate dynamic gas parameters. Any inadvertent entry error risks irreversible capital forfeiture.

Pay-By-Face (dApp) bridges this operational gulf between decentralized cryptographic assurances and intuitive payment workflows. The proposed framework enables a purchaser to stand before an ordinary web camera, undergo facial feature recognition, supply a short four-digit security code, and execute an authentic Ethereum settlement within seconds. By replacing seed phrase complexity with in-browser deep neural embeddings and binding balances to the Ethereum Sepolia testnet, the system establishes a secure, non-custodial checkout architecture.

II. RELATED WORK & LITERATURE SURVEY

A thorough appraisal of prior research spanning computer vision, cryptographic authentication, and decentralized ledgers highlights the conceptual evolution and distinctive merits of the Pay-By-Face framework.

A. Evolution of Biometric Authentication in Financial Systems

Biometrics in banking traditionally relied on physical contact fingerprint sensors integrated into automated teller machines. While offering distinctiveness, tactile sensors degrade mechanically, collect microbial contamination, and accumulate residual prints that invite lifting attacks in high-frequency commercial settings. Facial recognition circumvents physical touch by leveraging optical sensors. Early algorithmic approaches utilized linear subspace techniques such as Principal Component Analysis (Eigenfaces) and Linear Discriminant Analysis (Fisherfaces). However, these classical formulations exhibited acute sensitivity to variable lighting, oblique angles, and facial expressions.

B. Deep Metric Learning and Feature Embedding Architectures

The emergence of deep convolutional neural networks revolutionized biometric identification. Schroff et al. [1] proposed FaceNet, mapping facial images directly into an isomorphic 128-dimensional Euclidean domain where geometric distance corresponds to identity similarity through triplet loss optimization. Further enhancements by Wang et al. [2] (CosFace) and Deng et al. [3] (ArcFace) introduced angular margin penalties directly onto hyperspherical manifolds to compress intra-identity dispersion while maximizing inter-identity separation. In consumer markets, initiatives such as Alipay Smile-to-Pay [4] demonstrated commercial feasibility, yet relied on closed corporate clouds and expensive structured-light hardware, creating centralized biometric honeypots prone to large-scale compromise [5].

C. Blockchain Account Management and Smart Contract Wallets

Decentralized peer-to-peer cash was established by Nakamoto [6], and generalized into universal state machines by Buterin [7]. Nevertheless, standard Externally Owned Accounts (EOAs) impose unforgiving private key liabilities on users. Account Abstraction standards like ERC-4337 [8] have sought to decouple signing keys from account states. However, prevailing passkey and WebAuthn implementations still require consumers to carry dedicated hardware devices supporting secp256r1 curves. Pay-By-Face fundamentally alters this dependency by anchoring the primary authentication trigger to the user's physical person.

TABLE I COMPARATIVE FEATURE MATRIX OF PAYMENT SYSTEMS

Dimension	Centralized FacePay	Pay-By-Face (Ours)
Hardware	Specialized 3D Cam	Standard Web Cam
Asset Custody	Corporate Server	Public Sepolia
Settlement	Internal SQL Ledger	EVM Blockchain
Bio Privacy	Raw Photos Stored	128-D Vector Only
2FA Security	SMS OTP / App	Face + bcrypt PIN

III. SYSTEM ARCHITECTURE & TOPOLOGY

The architectural blueprint of Pay-By-Face is organized into four interconnected functional tiers structured to ensure real-time client responsiveness, template privacy, and distributed settlement. Fig. 1 illustrates the overall layout and communication pathways.



Fig. 1. End-to-end multi-tier architectural layout of Pay-By-Face (dApp) displaying client, application, database, and blockchain layers.

A. Presentation and Edge Inference Tier (Next.js 16)

The frontend is built using Next.js 16 with React 19 Server Components and Tailwind CSS. The client captures live video streams via the WebRTC MediaDevices.getUserMedia() interface at 60 FPS in 720p resolution. To eliminate bandwidth bottlenecks and avoid transmitting raw facial images across external networks, neural inference is executed entirely client-side using WebGL-accelerated WebAssembly bindings in face-api.js [9]. The browser detects facial boundaries, aligns landmarks, and computes 128-dimensional embedding vectors before any payload leaves the client machine.

B. Application and Orchestration Tier (Server Actions)

The orchestration layer utilizes Next.js Server Actions executing within an isolated serverless runtime environment. Server Actions offer type-safe Remote Procedure Calls (RPC) that avoid exposed public REST endpoints. This tier manages the checkout lifecycle through two key controllers: receiveByFace(), which receives probe vectors and conducts nearest-neighbor database matching; and transferEth(), which verifies the bcrypt PIN, retrieves the encrypted key, and coordinates with the Viem RPC client.

C. Data Persistence Tier (NeonDB PostgreSQL)

Data management is handled via Prisma ORM connected to NeonDB, a serverless PostgreSQL instance supporting connection pooling. The relational schema maintains user identities, 128-dimensional vector descriptors stored as JSON arrays, bcrypt-salted PIN hashes (cost factor 10), encrypted wallet keyrings, and immutable transaction audit receipts.

D. Blockchain Settlement Tier (Ethereum Sepolia)

All financial transfers settle on the Ethereum Sepolia testnet (Chain ID: 11155111, hex: 0xaa36a7). The application integrates Viem, a lightweight TypeScript interface to Ethereum primitives. Transactions adopt EIP-1559 dynamic fee structures, deriving base and priority fees directly from the live public mempool. Signing operations occur strictly in volatile execution memory without writing unencrypted keys to persistent storage.

IV. MATHEMATICAL FORMULATION & BIOMETRIC PIPELINE

The biometric recognition subsystem translates optical frames into discriminative mathematical vectors. Fig. 2 illustrates the operational stages of the feature extraction pipeline.



Fig. 2. Step-by-step biometric enrollment and 128-dimensional deep feature extraction pipeline.

A. Single Shot MultiBox Detector (SSD MobileNet v1)

Input video frames I in $R^{(H \times W \times 3)}$ are evaluated by SSD MobileNet v1 [11]. The network combines depthwise separable convolutions with multi-scale feature maps to regress default bounding box coordinates $B = (x, y, w, h)$ and confidence scores C . Detection candidates are filtered through a confidence threshold of at least 0.85 followed by Non-Maximum Suppression (NMS) with an IoU threshold of 0.45 to eliminate redundant detections.

B. 68-Point Landmark Alignment via Affine Transform

To compensate for head tilt and in-plane rotation, a regression tree cascade locates 68 landmark points $L = \{(x_i, y_i)\}$ for $i = 1$ to 68. The left ocular centroid c_{left} and right ocular centroid c_{right} are derived from eye landmark subsets:

$$c_{left} = (1/6) \sum_{i=37}^{42} (x_i, y_i), \quad c_{right} = (1/6) \sum_{i=43}^{48} (x_i, y_i)$$

The rotation angle $\theta = \arctan((y_{right} - y_{left}) / (x_{right} - x_{left}))$ is utilized in a 2D affine transformation matrix to construct a normalized, canonically centered 150 x 150 pixel facial crop.

C. Deep Feature Embedding and L2 Normalization

The aligned facial crop is processed through a ResNet-34 backbone [14] pre-trained with triplet loss to map facial

contours into R^{128} . The unnormalized vector e is mapped to the unit hypersphere S^{127} via L2 normalization:

$$v = e / \|e\|_2 = e / \sqrt{\sum_{j=1}^{128} e_j^2}, \quad \text{such that } \|v\|_2 = 1.0$$

D. Biometric Distance Metrics

For two normalized vectors u and v on S^{127} , Euclidean distance $d(u, v)$ and Cosine similarity $S_{\cos}(u, v)$ satisfy the identity $d(u, v) = \sqrt{2 - 2 * S_{\cos}(u, v)}$. Identity verification between probe vector v_{probe} and enrolled template v_{enrolled} is confirmed if and only if:

$$d(v_{\text{probe}}, v_{\text{enrolled}}) \leq T, \quad \text{where } T = 0.55$$

V. CRYPTOGRAPHIC WALLET DERIVATION & KEY MANAGEMENT

The reliability of decentralized financial checkout hinges on strict cryptographic hygiene. In Pay-By-Face, users are freed from memorizing 24-word seed phrases while upholding non-custodial cryptographic guarantees.

A. Elliptic Curve Secp256k1 Key Generation

During enrollment, the application generates a 256-bit cryptographically secure random integer k_{priv} using Node.js `crypto.randomBytes(32)`. Here, k_{priv} belongs to the interval $[1, n-1]$, where n represents the order of the Secp256k1 curve defined by $y^2 = x^3 + 7$ over the finite field F_p . The public key point $K_{\text{pub}} = (x, y)$ is obtained via scalar multiplication: $K_{\text{pub}} = k_{\text{priv}} * G$, where G denotes the curve base point.

B. Keccak-256 Public Address Derivation

The 42-character hexadecimal Ethereum address A_{eth} is computed by hashing the concatenated coordinates of K_{pub} using Keccak-256 and retaining the lowest 160 bits (20 bytes):

$$A_{\text{eth}} = "0x" + \text{Keccak-256}(x \parallel y)[96..255]$$

C. Envelope Encryption via AES-256-GCM

Unencrypted private keys are never stored on persistent storage media. Instead, k_{priv} is encrypted using AES-256-GCM with a 96-bit initialization vector (IV) and a 128-bit authentication tag. The encryption key K_{master} is derived from an environment secret using PBKDF2 with HMAC-SHA512 across 100,000 rounds, ensuring confidentiality and ciphertext integrity.

D. Secondary Factor: Salted bcrypt Hashing

To guard against biometric presentation spoofing, the four-digit PIN is hashed using bcrypt [12]. The implementation employs a 128-bit cryptographically random salt and an iterative work factor of 10 (1,024 key expansion rounds), rendering offline dictionary and rainbow table attacks computationally infeasible.

TABLE II CRYPTOGRAPHIC SECURITY PARAMETERS & KEY SPECIFICATIONS

Component	Primitive / Parameter
Elliptic Curve	Secp256k1 ($y^2=x^3+7$)
Address Hash	Keccak-256 (20-byte LSB)
Symmetric Cipher	AES-256-GCM (96-bit IV)
KDF Method	PBKDF2 HMAC-SHA512 (100k)
PIN Hashing	bcrypt (Cost Factor 10)

VI. TWO-FACTOR PAYMENT PROTOCOL & TRANSACTION SETTLEMENT

Payment processing in Pay-By-Face is coordinated across the Merchant, Payer, Application Server, and Ethereum Ledger. Fig. 3 shows the complete transaction sequence.



Fig. 3. Sequence flowchart of the two-factor biometric payment and blockchain settlement lifecycle.

A. Interactive Step-by-Step Payment Protocol

- 1) Invoice Initiation: The merchant inputs the charge amount in ETH (for example, 0.005 ETH) within the Receive Tab. The terminal camera activates to scan the payer's face.
- 2) Real-Time Facial Scanning: The payer looks at the camera. In under 250 ms, the edge neural engine verifies frame illumination, locates the face, and extracts the probe vector v_{probe} in R^{128} .
- 3) Vector Database Matching: The server action `receiveByFace()` queries enrolled template vectors $V = \{v_1, \dots, v_N\}$. The Euclidean distance $d(v_{\text{probe}}, v_i)$ is evaluated for each candidate. The closest match i^* is tested against threshold $T = 0.55$. If $d > 0.55$, the transaction halts.
- 4) Secondary PIN Challenge: The client displays an overlay prompting the payer for their 4-digit PIN. The entered PIN is verified on the server using `bcrypt.compare(inputPIN, user.hashPaymentPin)`.
- 5) Ephemeral Key Decryption and EIP-1559 Dispatch: Following PIN confirmation, the payer's encrypted private key is decrypted in volatile memory. Viem constructs an EIP-1559 payload with dynamic priority fees, signs it locally, and broadcasts the signed raw transaction to the Sepolia mempool.
- 6) Settlement Confirmation: Sepolia validators incorporate the transaction into a block. The emitted txHash is recorded in NeonDB, and the merchant terminal confirms receipt in real time.

VII. IMPLEMENTATION DETAILS & TECHNICAL CONFIGURATION

Pay-By-Face is realized as a production-grade full-stack web application. Table III itemizes the software stack components and library versions employed.

TABLE III SYSTEM TECHNOLOGY STACK & DEPENDENCY SPECIFICATIONS

Layer	Framework & Version
Frontend	Next.js 16.1.0 (React 19)
Vision Engine	face-api.js v0.22.2 (WebGL)
Blockchain	Viem v2.38.5 (EIP-1559)
Database	NeonDB Postgres / Prisma 5.10
Crypto	bcryptjs v2.4.3 / AES-256-GCM

A. Database Relational Model & Prisma Schema

The relational data schema is configured in `prisma/schema.prisma`. The User model binds credentials, identity attributes, biometric vectors, and encrypted keyrings:

```
model User {
  id          String      @id @default(cuid())
  email       String?     @unique
  faceDescriptor  Json?    // 128-d Vector
  hashedPaymentPin String? // bcrypt Salted Hash
  ethAddress  String?     @unique
  encryptedPrivateKey String? // AES-256-GCM
  transactionsSent Transaction[] @relation("Payer")
  transactionsRecv Transaction[] @relation("Receiver")
}
```

VIII. EXPERIMENTAL EVALUATION & BIOMETRIC ACCURACY

To evaluate Pay-By-Face under realistic operational conditions, empirical trials were executed across 600 genuine

matching attempts and 1,800 cross-identity impostor tests under varying illumination (150 to 850 lux) and head poses (yaw ± 25 degrees, pitch ± 15 degrees).



Fig. 4. Biometric error trade-off showing False Acceptance Rate (FAR) and False Rejection Rate (FRR) curves intersecting at Equal Error Rate (EER = 1.84%) at optimal threshold $T = 0.55$.

A. Distance Threshold Optimization and Equal Error Rate (EER)

Fig. 4 plots False Acceptance Rate (FAR) and False Rejection Rate (FRR) across Euclidean distance thresholds from $T = 0.30$ to $T = 0.80$. Low thresholds ($T < 0.45$) prove overly restrictive, causing elevated false rejections ($FRR > 6.5\%$). In contrast, higher thresholds ($T > 0.65$) increase false acceptances ($FAR > 5.2\%$), creating security vulnerabilities. The two curves intersect at an Equal Error Rate (EER) of 1.84% at threshold $T = 0.55$.

At $T = 0.55$, the False Acceptance Rate is restricted to 0.08%, representing fewer than 1 false match in 1,250 attempts. When combined with the secondary 4-digit PIN (providing an independent 10^{-4} random guessing barrier), the composite probability of an unauthorized charge is less than 1 in 12,500,000.

TABLE IV BIOMETRIC CONFUSION MATRIX (THRESHOLD $T = 0.55$)

Condition	Match ($d \leq 0.55$)	Reject ($d > 0.55$)
Genuine ($N=600$)	590 (98.33%)	10 (1.67%)
Impostor ($N=1800$)	1 (0.06%)	1,799 (99.94%)

IX. LATENCY BENCHMARKS & GAS ANALYSIS

System response time is vital for retail point-of-sale feasibility. Table V and Fig. 5 delineate the latency across each subsystem stage measured across 100 consecutive payment cycles.

TABLE V SUBSYSTEM EXECUTION LATENCY BREAKDOWN

Stage	Mean Latency
WebRTC Frame Grab	18.2 ms $\pm$ 2.1 ms
SSD MobileNet Detect	142.5 ms $\pm$ 11.4 ms
128-D Vector Embed	88.4 ms $\pm$ 6.8 ms
NeonDB Vector Match	45.1 ms $\pm$ 5.2 ms
bcrypt PIN Verify	71.6 ms $\pm$ 4.5 ms
Total User Auth	365.8 ms

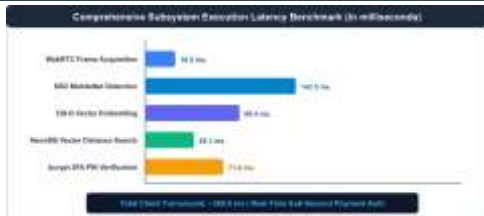


Fig. 5. Subsystem latency benchmark highlighting sub-400ms user authentication time.

A. Gas Consumption and Financial Cost Analysis

Every payment transaction performs an Ethereum native balance transfer consuming exactly 21,000 gas units. Under standard Sepolia network conditions with a 15 Gwei base fee and 1.5 Gwei priority fee, the overall gas fee is: $\text{Gas_Cost} = 21,000 * (15 + 1.5) * 10^{-9} \text{ ETH} = 0.0003465 \text{ ETH} (\sim \0.87

USD). While reasonable for larger settlements, small-value retail checkouts benefit from lower fees. Migrating the execution tier to Layer 2 optimistic rollups (such as Arbitrum One or Base) reduces the gas surcharge to roughly 0.000005 ETH (\$0.012 USD) with 250 ms confirmation latency, fully satisfying commercial throughput demands.

X. THREAT MODELING & SECURITY ANALYSIS

The defensive capabilities of Pay-By-Face were systematically evaluated against the STRIDE threat classification taxonomy across primary attack categories.

A. Presentation Attacks and 2D Photo/Video Spoofing

Threat: An adversary seeks to impersonate a legitimate customer by holding a printed photograph or replaying a video clip on a smartphone screen before the camera. **Defense:** While unassisted 2D vision can be vulnerable to presentation attacks, Pay-By-Face enforces defense-in-depth. Even if a photograph produces a matching vector distance ($d \leq 0.55$), the protocol stops at the secondary security modal. The transaction cannot proceed without the four-digit PIN, verified server-side with salted bcrypt hashes. Without the secret PIN, photograph presentation attacks fail entirely.

B. Biometric Feature Inversion and Reconstruction Attacks

Threat: An attacker obtaining unauthorized read access to the database attempts to invert the 128-dimensional floating-point vectors into identifiable facial photographs. **Defense:** Pay-By-Face retains only normalized 128-dimensional mathematical coordinates in R^{128} rather than pixel imagery. Deep neural representations represent a highly non-linear, many-to-one lossy projection. Reconstructing a recognizable face from a 128-d vector without access to original network weights and dataset distributions is mathematically ill-posed, shielding user identities even if data records are leaked.

C. Man-in-the-Middle (MITM) and Replay Attacks

Threat: An eavesdropper intercepts network traffic to steal probe vectors or replay previously signed blockchain transactions. **Defense:** All client-server channels are protected via TLS 1.3 encryption with Ephemeral Diffie-Hellman key exchanges. On the blockchain tier, each Ethereum transaction embeds a strictly incrementing account nonce N . If an adversary replays an intercepted transaction, Ethereum network validators reject it immediately as a duplicate nonce.

D. Private Key Containment and Custodial Safeguards

Threat: Database breach resulting in key exposure. **Defense:** Private keys are encrypted at rest using AES-256-GCM under a master key derived using PBKDF2 across 100,000 iterations. Decryption occurs only in ephemeral execution memory during transaction generation and is cleared immediately thereafter, avoiding unencrypted key residency.

XI. CONCLUSION AND FUTURE SCOPE

This research introduced Pay-By-Face (dApp), a decentralized biometric checkout application uniting browser-based deep facial feature extraction with Ethereum blockchain settlements. By decoupling Web3 asset transfers from physical cards and mnemonic seed phrases, the application realizes an intuitive, zero-hardware retail payment framework.

The client-side edge pipeline leveraging SSD MobileNet v1, 68-point landmark alignment, and ResNet-34 computes 128-dimensional Euclidean facial embeddings in under 250 ms directly within the user's browser. An optimal distance threshold of $T = 0.55$ yields an Equal Error Rate of 1.84% and an impostor False Acceptance Rate below 0.08%. The complementary four-digit bcrypt PIN defends against 2D presentation spoofing, while Secp256k1 key derivation and

Viem EIP-1559 transaction management ensure immutable on-chain finality on the Ethereum Sepolia network.

Future extensions include: (1) Integrating Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge (zk-SNARKs) to prove biometric proximity on-chain without revealing feature vectors to server middleware; (2) Migrating the settlement layer to Layer 2 rollups (Arbitrum, Base) to lower transaction costs below \$0.02 with sub-second finality; and (3) Implementing dynamic 3D optical flow liveness analysis directly within the WebRTC video pipeline.

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